Stable cursor decoding across months and a closed-loop WebGrid simulator for intracortical BCIs

Amal Mehta*

October 2026

Abstract

A cursor brain–computer interface (BCI) is only as useful as its worst day: people with paralysis need control that works when they sit down, without a daily calibration block, and what they experience is throughput, not offline R^2. We study both problems on 24 public sessions from a 96-electrode motor-cortex array spanning four months (monkey Indy, random-target cursor reaching). We compare Wiener, Kalman, factor-analysis-stabilised and GRU decoders with a *Stable Latent Decoder* (SLD): identity-initialised per-day channel adapters feeding a shared recurrent latent model, trained on five calibration days with input perturbations and a masked-spike loss, and adapted to a new day without labels (NoMAD-style likelihood plus latent-moment matching) or with a few labelled reaches. Within a day, SLD and the GRU are tied ($R^2 = 0.791$ vs 0.788; Wiener 0.528). On 17 held-out days up to 138 days later, with no labels, re-centred SLD keeps $R^2 = 0.718$ (GRU 0.662, Wiener 0.422, fixed Wiener -0.037). The ablations show that multi-day training, input perturbation and per-channel re-centring do most of the work; latent alignment helps only far from calibration and on two sessions where electrode identities changed, which a label-free firing-rate check flags. With half the electrodes silenced SLD keeps $R^2 = 0.689$ against 0.393 for the GRU, and adapter-only few-shot recalibration is stable with a single labelled reach, where fine-tuning the whole GRU is not. A closed-loop WebGrid simulator, with a feedback-control user and a neural encoder fit to each day, shows that offline rankings do not transfer: on a 10×10 grid the GRU reaches 1.66 bits/s, SLD 1.35, the Wiener filter 1.27 (2.43 after ReFIT-style recalibration) and the Kalman filter 0.00, against a 4.70 bits/s perfect-decoder ceiling. All bitrates are simulated, not measured in people or animals.

1 Introduction

Neuralink's Telepathy product lets a person with paralysis control a computer cursor by thought. Its public scorecard is Webgrid: click targets on a grid as fast as possible; the score is an achieved bitrate in bits per second (BPS) [1]. Three machine-learning problems decide that number. The decoder must turn spiking into cursor velocity with low latency; it must keep working as the recorded neurons change from day to day (Neuralink's first participant lost signal when threads retracted, and recovered performance by changing the recording and decoding algorithms [2]); and it must be judged in closed loop, where the user corrects errors in real time and offline accuracy can mislead [3–5].

This report builds and evaluates a small, complete stack for those three problems on public data. Contributions:

1. A cross-day evaluation in the FALCON style [6] on 24 sessions (5 calibration, 19 held-out over four months) of the O'Doherty random-target dataset [7], with zero-label, few-shot and electrode-failure conditions and strong classical baselines.
2. The Stable Latent Decoder: day-specific identity-initialised channel adapters and a shared causal latent model, trained with augmentation [8] and masked-spike reconstruction [9], adapted by a NoMAD-style objective [10] or by adapter-only few-shot learning.
3. A closed-loop WebGrid simulator whose neural encoder is fit to each day and whose noise is calibrated so that open-loop decoding accuracy matches real data, with per-decoder tuning of gain, smoothing and click caution, a perfect-decoder ceiling, a grid-size sweep and a ReFIT-style closed-loop recalibration.
4. An honest negative result and a practical rule: on these data, re-centring each electrode's statistics captures most of the benefit latent alignment is meant to provide; alignment matters when electrode identities change, which a label-free check can detect.

2 Background

Signals and decoders. Clinical intracortical BCIs decode per-electrode threshold crossings binned at 10–50 ms, without spike sorting. Cursor control decodes *velocity*; the Kalman filter [11] and its closed-loop recalibrated variant ReFIT-KF [3] long set the standard, and neural-network decoders now beat them online [12].

*Independent project; not affiliated with or endorsed by Neuralink.

Grid-task bitrate [13]

$$B = \frac{\log_2(N-1)\ \max(S_c - S_i, 0)}{t} \tag{1}$$

(N targets, S_c/S_i correct/incorrect selections in t seconds) is the field's throughput metric; people with BrainGate arrays reached 1.4–3.7 bits/s [14], and Neuralink reports participants around 8–10+ BPS on Webgrid.

Nonstationarity. Units drift, appear and vanish across days. Latent population dynamics stay stable [15], which motivates aligning each day into a fixed latent space: factor-analysis alignment of stable channels [16], adversarial alignment [17], and NoMAD, which freezes an LFADS dynamics model and learns an alignment network by maximising spike likelihood while matching latent distributions [10, 18]. A cheaper alternative tracks feature means and recalibrates on task data [19]; pretrained multi-session transformers adapt with few labels [20–22]; pseudo-label recalibration has kept a handwriting BCI stable for a year [23]. FALCON standardises these comparisons [6].

3 Data

We use 24 sessions from monkey Indy in the O'Doherty et al. dataset [7, 24] (Zenodo 583331), recorded from 15 Sept 2016 to 31 Jan 2017 with a 96-electrode Utah array in primary motor cortex while the monkey moved a cursor between random targets with its arm (the session `indy_20161005_06` is NLB's MC_RTT [25]). For each electrode we merge all sorted units and unsorted threshold crossings into one count per 20 ms bin. Cursor velocity is the derivative of the Gaussian-smoothed (40 ms) cursor position. The decoding target at bin t is velocity at $t + \ell$; $\ell = 20$ ms was selected on calibration-day validation data. Each session lasts 6–14 minutes (300–640 reaches). The first five sessions (days 0–15) are calibration days; the other 19 (days 20–138) are held out. Within every session the first 60 % is the train block, the next 10 % validation and the last 30 % test. On a held-out day the train block is used only as *unlabelled* spikes, except for the first N reaches in the few-shot condition.

4 Methods

4.1 Baselines

Wiener filter: ridge regression from 10 bins (200 ms) of z-scored counts to velocity. **Kalman filter**: velocity state, linear-Gaussian observation of causally smoothed counts, steady-state gain; smoothing window and an observation-noise scale tuned on validation. Both are fit per day (within-day reference) and on all five calibration days (the deployed version). **FA stabiliser** after Degenhart et al. [16]: 10-factor FA on calibration days, ridge decoder on lagged factors; on a new day FA is refit without labels, the 60 % of channels whose loadings agree best are found iteratively, and an orthogonal Procrustes map aligns the new factors. **GRU**: input layer, one GRU layer (128 units), linear readout, trained on all calibration days.

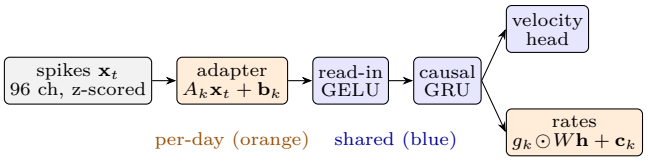


Figure 1: Stable Latent Decoder. Only the adapter $(A_k, \mathbf{b}_k)$ and the rate gain/offset $(g_k, \mathbf{c}_k)$ are specific to day k. A new day starts from $A = I$, i.e. from the re-centred decoder.

Re-centring. Every decoder normalises each channel by the mean and standard deviation of the current day. A *fixed* decoder keeps the statistics of the last calibration day; a *re-centred* decoder recomputes them from the new day's unlabelled spikes [19]. This costs nothing and needs no labels, so we give it to every baseline.

4.2 Stable Latent Decoder

For day k with z-scored counts $\mathbf{x}_t \in \mathbb{R}^C$ (Fig. 1):

$$\mathbf{u}_t = \mathrm{GELU}\big(W_{\mathrm{in}}(A_k\mathbf{x}_t + \mathbf{b}_k)\big), \tag{2}$$

$$\mathbf{h}_t = \mathrm{LN}\big(\mathbf{u}_t + \mathrm{GRU}(\mathbf{u}_{\le t})\big), \tag{3}$$

$$\hat{\mathbf{v}}_t = W_v\mathbf{h}_t + \mathbf{b}_v, \qquad \log \boldsymbol{\lambda}_t = \mathbf{g}_k \odot (W_r\mathbf{h}_t) + \mathbf{c}_k. \tag{4}$$

$A_k \in \mathbb{R}^{C\times C}$ is initialised to the identity. Training on the calibration days minimises

$$\mathcal{L} = \|\hat{\mathbf{v}} - \mathbf{v}\|^2 + \beta\,\mathrm{NLL}_{\mathrm{Pois}}(\mathbf{n}_M \mid \boldsymbol{\lambda}_M) + \gamma \textstyle\sum_k \big(\|A_k - I\|_F^2 + \|\mathbf{b}_k\|^2\big), \tag{5}$$

where on half of each batch a random 20 % of channels M are zeroed at the input and their raw counts $\mathbf{n}_M$ must be predicted from the others (masked-channel reconstruction, $\beta = 0.5$, $\gamma = 0.01$). Inputs are perturbed as in Sussillo et al. [8]: per-channel gain and offset jitter ($\sigma = 0.1$) and silencing of a random 0–15 % of channels. Training uses 64-bin windows (first 24 bins are context), AdamW with a one-cycle schedule, and model selection on validation R^2. A banded-attention causal transformer with ALiBi position bias is available as an alternative backbone.

Unsupervised alignment. On a new day we add a slot k^* with $A = I$ and freeze everything except $(A_{k^*}, \mathbf{b}_{k^*}, \mathbf{g}_{k^*}, \mathbf{c}_{k^*})$. Following NoMAD [10], we minimise the masked-spike Poisson NLL of the new day's unlabelled spikes through the frozen dynamics, plus $\eta\,\mathrm{KL}\big(\mathcal{N}(\boldsymbol{\mu}_{k^*}, \Sigma_{k^*}) \,\|\, \mathcal{N}(\boldsymbol{\mu}_0, \Sigma_0)\big)$ between Gaussian fits

to the new day's latents $\mathbf{h}$ and to the calibration days' latents ($\eta = 0.1$), plus the identity penalty; 300 Adam steps.

Few-shot recalibration. With N labelled reaches we fit the same day-specific parameters to the velocity loss (adapter only, 150 steps), optionally together with the unsupervised terms after alignment. Baselines: a Wiener filter fit from scratch on the N reaches; a Wiener filter whose ridge penalty shrinks toward the calibrated weights, $\|Y - ZW\|^2 + \alpha\|W - W_0\|^2$; and fine-tuning all GRU weights.

4.3 Closed-loop WebGrid simulator

Encoding model. For each simulated day we fit an MLP (two hidden layers of 128) from intended velocity and speed over a window of -100 to $+140$ ms to log firing rates of all 96 channels, by Poisson likelihood on the train block. The window extends into the "future" of the decoding target because motor cortex leads movement. Arm position is not an input: a paralysed user has no arm posture signal. Shared variability is added in log-rate space as a rank-4 AR(1) process ($\tau = 200$ ms) with loadings from factor analysis of smoothed residuals; its scale is chosen from $\{0, 0.25, \ldots, 1\}$ so that the Wiener filter's R^2 on simulated spikes, generated from the real test-block velocities, matches its real test R^2.

User model. Intended velocity follows a delayed feedback controller in the spirit of optimal-feedback-control user models [5]. The user sees the cursor 100 ms late, predicts its present position with an efference copy of its own recent intentions, and commands $\mathbf{v}^{\mathrm{cmd}} = v_{\max}(1 - e^{-d/d_0})\,\hat{\mathbf{e}}$ toward the target centre (d the predicted distance; zero inside 0.35 cell). $v_{\max} = 80$ mm/s and $d_0 = 16$ mm are fit to the monkey's own reaches. The command passes a first-order 80 ms motor lag and 10 % signal-dependent noise.

Task. A 130×130 mm workspace (the monkey's reach range) is divided into an $n \times n$ grid; one random cell is the target. The user clicks when the delayed view and the prediction both place the cursor inside the target for a hold time and the cursor looks slow; the click registers 40 ms later, is missed with probability 0.05, and spurious clicks occur at 0.05/s. A correct selection spawns a new target. Bitrate follows Eq. 1 over 60 s. For each decoder, the cursor gain $\{0.75, 1, 1.5\}$, exponential velocity smoothing $\{0, 0.6\}$ and the user's hold time $\{0.1, 0.25\}$ s are tuned on two separate 30 s sessions and frozen. The oracle moves the cursor with the true intention at the same latency as a decoder.

ReFIT in simulation. For the linear decoders we also run 120 s of closed-loop use, relabel the intended velocity by rotating the decoded velocity toward the target (zero on the target) [3], refit, re-tune and re-evaluate.

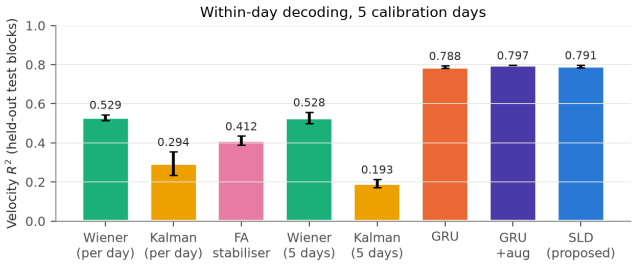


Figure 2: Within-day velocity R^2 on the calibration days' test blocks (error bars: sd over calibration days or seeds).

5 Results

5.1 Within-day decoding

Table 1 and Fig. 2 show that both recurrent decoders are far ahead of the linear ones: SLD reaches $R^2 = 0.791$ and the GRU 0.788, against 0.528 for a Wiener filter trained on the same five days. The steady-state velocity Kalman filter is weak at 20 ms bins: its smoothing either lags the movement or passes noise, and the validation-tuned compromise stays below the Wiener filter (Section 6). Within one day, SLD and the GRU are statistically indistinguishable (seed sd $\approx$0.007); the proposed parts of SLD are not meant to win here.

5.2 Across days without labels

Fig. 3 tracks every decoder over four months. A fixed Wiener filter is unusable (mean $R^2 = -0.130$; individual days fall below -1), and even the recurrent decoders lose roughly half their accuracy when left fixed (GRU 0.299, SLD 0.388). Re-centring each channel with the new day's unlabelled statistics recovers most of the loss. On the 17 days without an electrode re-mapping, re-centred SLD averages $R^2 = 0.718$ and the re-centred GRU 0.662, close to their within-day accuracy, while the Wiener filter reaches 0.422 and the FA stabiliser 0.305.

What makes a decoder hold up? The ablation (Table 2) separates the ingredients. Training with Sussillo-style input perturbations is worth about $+0.05$ R^2 across days for the GRU (0.662 $\rightarrow$ 0.712), and SLD inherits it; removing augmentation from SLD costs less across days (0.701 vs 0.718), possibly because the masked-spike loss also exposes it to missing channels. The per-day adapters and the masked-spike loss make no measurable difference to re-centred accuracy (0.725 without adapters, 0.711 without the loss), and the banded transformer backbone is slightly worse than the GRU (0.698).

Table 1: Main results. Within-day: velocity R^2 on the held-out last 30 % of each calibration day, mean ± sd over 3 seeds (GRU, SLD) or over the 5 calibration days (linear decoders). Cross-day: mean ± sd over the 19 held-out days (days 20–138), with no labels; "fixed" keeps the last calibration day's normalisation; "adapted" is re-centring (Wiener, Kalman, GRU, SLD), FA/Procrustes alignment, or SLD's unsupervised alignment. Bitrate: simulated WebGrid, click selection, mean ± sd over 4 sessions of 60 s after per-decoder tuning. **All bitrates come from the simulator, not from a person or animal.**

Decoder	Within-day R^2	Cross-day R^2 (19 days, no labels)		Sim. bits/s
		fixed	adapted	(10×10)
Wiener filter (per day)	0.529 ± 0.015	–	–	–
Kalman filter (per day)	0.294 ± 0.061	–	–	–
Wiener filter (5 days)	0.528 ± 0.030	-0.130 ± 0.575	0.353 ± 0.207	1.27 ± 0.37
Kalman filter (5 days)	0.193 ± 0.021	-1.894 ± 2.417	0.064 ± 0.225	0.00 ± 0.00
FA stabiliser [16]	0.412 ± 0.022	0.086 ± 0.338	0.235 ± 0.212	–
GRU	0.788 ± 0.006	0.299 ± 0.154	0.589 ± 0.223	1.66 ± 0.11
SLD (proposed), re-centred	0.791 ± 0.007	0.388 ± 0.183	0.636 ± 0.244	1.35 ± 0.18
SLD (proposed), aligned	–	–	0.603 ± 0.349	–
Oracle (perfect decoder)	–	–	–	4.70 ± 0.17

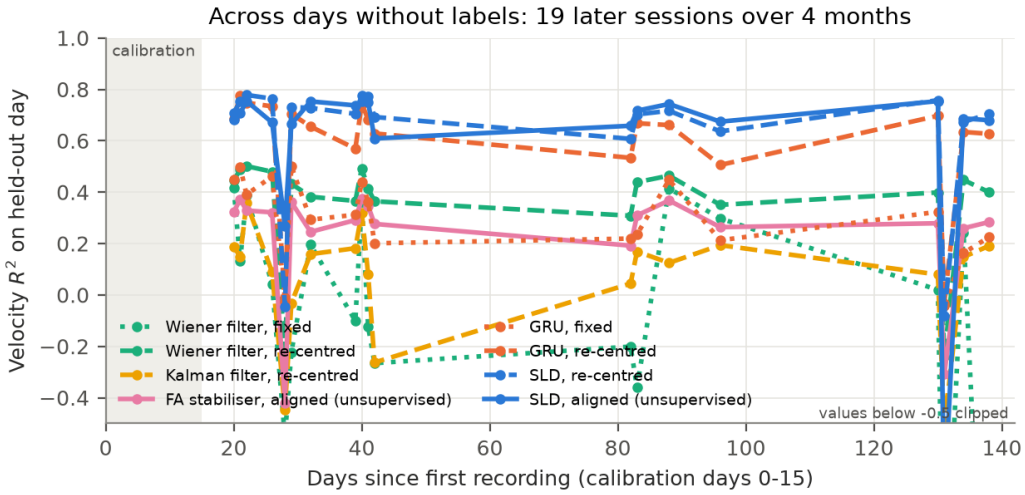


Figure 3: Velocity R^2 on each held-out day, no labels. Dotted: fixed decoder; dashed: re-centred; solid: unsupervised alignment. Days 28, 131 are electrode re-mapping events (Sec. 5.3); every re-centred decoder collapses there.

Does unsupervised alignment help? Mostly not, on natural drift. Averaged over the 17 regular days, aligned SLD (0.706) is no better than re-centred SLD (0.718). The effect depends on elapsed time: within six weeks of calibration alignment costs -0.033 R^2 on average, while 82–138 days out it gains +0.016. Starting from the re-centred decoder and penalising departures from the identity limits the damage, but the objective (spike likelihood plus latent-moment matching) is not tied to velocity, so when the re-centred decoder is already good it can only move away from it.

5.3 Electrode re-mapping events

Two held-out sessions (days 28, 131) break every re-centred decoder ($R^2 \approx 0$). A decoder trained on that day alone still reaches $R^2 > 0.5$, so the neurons still encode velocity; what changed is which electrode records what. A label-free check confirms it: the correlation between each channel's mean firing rate on the new day and on the last calibration day is -0.24, -0.17 on these two days and between 0.35 and 0.77 on all other held-out days. This is the situation latent alignment is designed for. On day 28, SLD alignment recovers $R^2 =$ 0.568, 0.533, -0.300 (three seeds) from -0.045 re-centred; on day 131 it recovers one seed (0.438, -1.051, -1.816) and makes the others worse. Alignment can find the re-mapping but is not reliable; in a product it should be gated by the detector above and verified before it is trusted.

5.4 Electrode failure and re-wiring

Fig. 4 silences or scrambles a growing fraction of the 96 channels. Silencing is where augmentation matters most. With half the electrodes dead, the plain GRU falls to $R^2 = 0.063$ (0.393 after re-centring), the augmented GRU to 0.484, while SLD keeps 0.689. Already at 10 %

Table 2: Ablations (velocity R^2). Within-day: test blocks of the calibration days. Cross-day means exclude the two re-mapping days; aligned variants of single-seed ablations were run on a 7-day subset (days 20, 26, 32, 41, 83, 130, 138), so the last two columns compare all models on that subset.

Model	Within-day	Cross-day, 17 days		Cross-day, 7-day subset	
		fixed	re-centred	aligned	re-centred
SLD (full)	0.791 ± 0.007	0.433	0.718	0.715	0.734
w/o input augmentation	0.805	0.300	0.701	0.727	0.716
w/o masked-spike loss	0.784	0.447	0.711	0.702	0.727
w/o day adapters	0.796	0.434	0.725	–	0.736
transformer backbone	0.764	0.456	0.698	0.671	0.713
GRU baseline	0.788 ± 0.006	0.337	0.662	–	0.679
GRU + augmentation	0.797	0.444	0.712	–	0.726

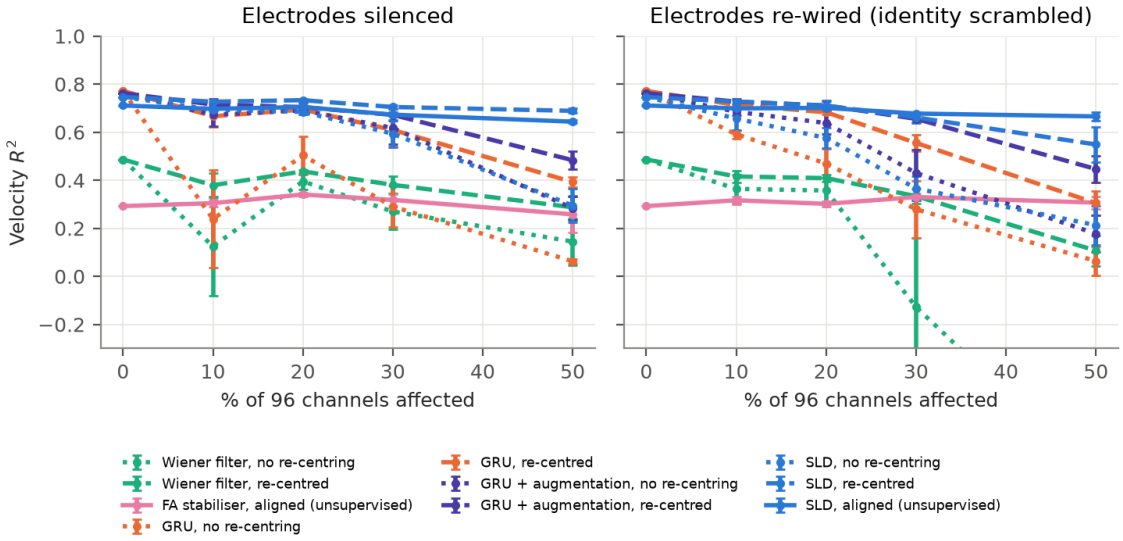


Figure 4: Stress tests on the last calibration day: a growing fraction of electrodes silenced (left) or swapped with each other (right); mean ± sd over 3 random draws.

silenced the plain GRU without re-centring drops to 0.238. For re-wired electrodes, re-centring again does most of the work; unsupervised alignment adds a little at 30 % re-wired (0.678 vs 0.661) and nothing at small fractions.

5.5 Few-shot recalibration

With labels, the question becomes how few reaches are needed (Fig. 5, Table 3). A Wiener filter trained from scratch is useless with 1–5 reaches and still poor at 50. Shrinking toward the calibrated weights is safe but slow. Fine-tuning every GRU weight on one reach damages the decoder (R^2 0.380 vs 0.468 at $N = 0$). Fitting only SLD's day adapter is stable at every N (0.511 at $N = 1$, 0.696 at $N = 50$), and starting it from the unsupervised alignment gives the best curve (0.719 at $N = 20$), largely because it rescues the re-mapping day in the sample.

5.6 Closed loop: simulated WebGrid

Fidelity. With the noise scale calibrated on the Wiener filter, the simulated spikes reproduce the linear decoders' accuracy (Wiener 0.487 real vs 0.438 simulated) and SLD's (0.746 vs 0.717), but the GRU loses more in simulation (0.772 vs 0.597): the encoder does not reproduce every feature the GRU relies on (Fig. 6).

Bitrate. Fig. 7 and Table 4 give the main closed-loop result. A perfect decoder at the same latency scores 4.70 bits/s on the 10×10 grid, the ceiling set by the user model and the task. The GRU scores 1.66, SLD 1.35, the Wiener filter 1.27 and the Kalman filter 0.00. Offline ranking does not carry over cleanly: SLD and the GRU are tied offline and the GRU scores slightly higher in closed loop, while the much less accurate Wiener filter comes close to SLD once its output is smoothed. The spread between the SLD ablations is as large as the differences between decoders (2.07 bits/s without adapters, 1.96 without augmentation, 0.99 without the masked-

Table 3: Few-shot recalibration (R^2, mean ± sd over 3 held-out days: 20, 28 and 40; one seed; CPU budget).

Method	$N = 0$	$N = 1$	$N = 5$	$N = 20$	$N = 50$
Wiener, from scratch	–	-0.08 ± 0.04	0.08 ± 0.01	0.28 ± 0.03	0.41 ± 0.05
Wiener, prior-centred ridge	0.24 ± 0.33	0.23 ± 0.33	0.30 ± 0.25	0.39 ± 0.14	0.45 ± 0.08
GRU, fine-tune all	0.47 ± 0.36	0.38 ± 0.24	0.51 ± 0.26	0.62 ± 0.17	0.68 ± 0.11
SLD, adapter only	0.49 ± 0.36	0.51 ± 0.32	0.57 ± 0.24	0.66 ± 0.12	0.70 ± 0.07
SLD, align + adapter	0.68 ± 0.08	0.68 ± 0.07	0.69 ± 0.07	0.72 ± 0.06	0.73 ± 0.05

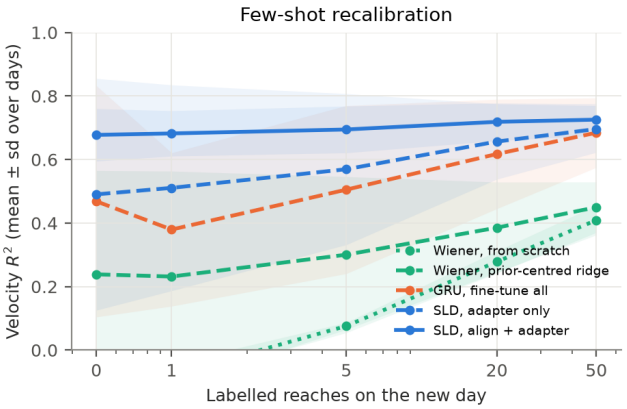


Figure 5: Few-shot recalibration on 3 held-out days (days 20, 28 and 40; day 28 is a re-mapping day; one seed). $N = 0$ is the label-free starting point of each method.

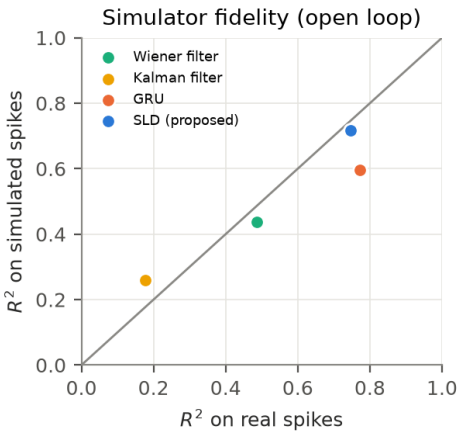


Figure 6: Simulator fidelity: each decoder's R^2 on real test spikes vs on simulated spikes generated from the same velocities.

spike loss, 0.99 with the transformer; GRU with augmentation 1.33), so differences below about 0.5 bits/s should not be read as real with four 60 s sessions. What is robust: every neural decoder beats the Kalman filter by a wide margin, the best decoder reaches only about a third of the perfect-decoder ceiling (1.66 vs 4.70 bits/s), and precision, not speed, limits them, because bitrate falls with grid size for every decoder while it rises for the oracle (5.81 bits/s for the oracle on 35×35, 0.43 for the GRU). Dwell selection is much worse than clicking for the decoders (Table 4): holding the cursor still on one small cell for 0.5 s is hard for a noisy cursor.

ReFIT in simulation. Recalibrating the Wiener filter on 120 s of its own closed-loop data with intention relabelling lifts it from 1.27 to 2.43 bits/s, the qualitative effect Gilja et al. reported in monkeys [3]. That the simulator reproduces a known closed-loop effect that offline metrics cannot show is the best evidence that it is useful. The same procedure does not rescue the Kalman filter (0.06 bits/s), whose problem here is its offline fit, not intention labels.

Across days in closed loop. Fig. 9 repeats the closed-loop test on four later days, with an encoding model fit to each day and no re-tuning. Fixed decoders fail outright (mean 0.14 bits/s for the fixed GRU, 0.00 for fixed SLD), confirming in closed loop what the offline numbers show. Re-centring restores the Wiener filter (1.36 bits/s on average) and the GRU (1.14). For SLD the closed loop reverses the offline picture: with a fresh identity adapter and re-centring it scores only 0.41 bits/s, while the unsupervised-aligned adapter scores 1.21, although the two differ little in offline R^2. The re-centred Wiener filter is the most consistent decoder across the four days in closed loop; the neural decoders degrade on the last day (day 138). Four days and three sessions per point are too few for firm rankings, but the pattern is a second warning against choosing a stabilisation method on offline R^2 alone.

Table 4: Closed-loop details: tuned gain, velocity smoothing and click hold time per decoder; simulated bits/s with click and with 0.5 s dwell selection (10×10).

Decoder	gain	smooth	hold (s)	click bits/s	dwell bits/s
Oracle	1	0	0.1	4.70 ± 0.17	4.20 ± 0.21
Wiener	0.75	0.6	0.1	1.27 ± 0.37	0.55 ± 0.45
Kalman	1	0	0.1	0.00 ± 0.00	0.00 ± 0.00
GRU	1	0	0.25	1.66 ± 0.11	0.00 ± 0.00
SLD	1.5	0.6	0.1	1.35 ± 0.18	0.00 ± 0.00
Wiener + ReFIT	1	0.6	0.1	2.43 ± 0.37	–
Kalman + ReFIT	0.75	0.6	0.1	0.06 ± 0.06	–

5.7 Latency

All decoders are cheap enough for a 20 ms update on one CPU thread (Table 5): the GRU and SLD need about

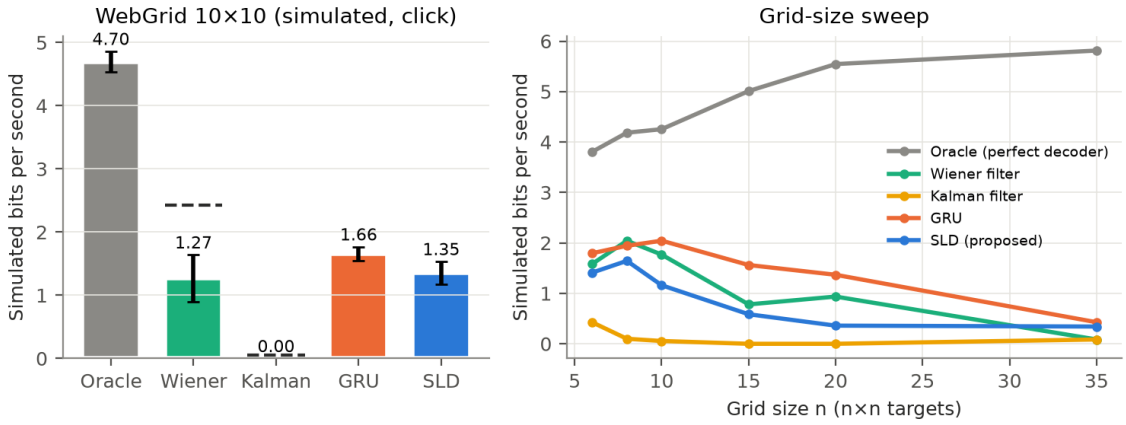


Figure 7: Simulated WebGrid bitrate. Left: 10×10 grid, 4 sessions of 60 s per decoder; dashed bars mark the linear decoders after ReFIT-style recalibration. Right: bitrate vs grid size (30 s sessions).

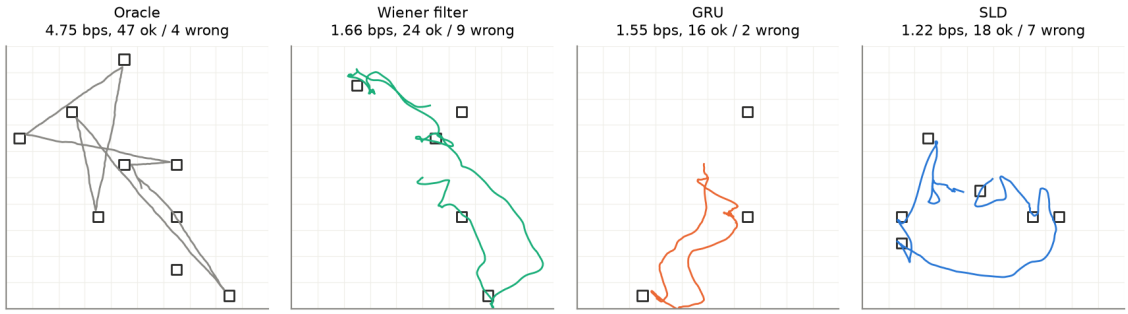


Figure 8: First 10 s of one simulated session per decoder (squares: targets).

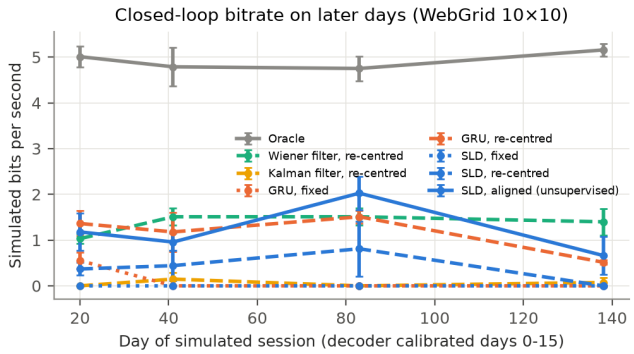


Figure 9: Simulated bitrate on later days with the decoders' day-0 tuning (3 sessions per point).

1 ms per bin, the banded transformer about 5 ms because it re-reads its context window at every step.

6 Discussion

Cheap things first. The largest cross-day gains here come from three inexpensive choices, not from the alignment machinery: training on several days, perturbing inputs during training [8], and re-centring each electrode's

Table 5: Streaming inference cost per 20 ms bin (one CPU thread, shared machine).

Decoder	ms per 20 ms bin
Wiener filter	0.073
Kalman filter	0.117
FA stabiliser	2.339
GRU	0.758
SLD (GRU backbone)	1.109
SLD (transformer backbone)	5.541

statistics on every new day without labels [19]. Together they keep a recurrent decoder within a few hundredths of its within-day R^2 for four months on this array. A practical consequence for a product like Telepathy is that the default "no-calibration" path should be a multi-day, augmented decoder with automatic re-centring; latent alignment is a second line of defence.

When alignment earns its keep. Alignment pays off when the electrode-to-neuron mapping changes: on the two re-mapping days and, slightly, more than 80 days after calibration. Neuralink's experience with thread retraction [2] is exactly a large mapping change; our results suggest pairing a label-free change detector (the firing-rate correlation used here flags both events with a wide margin) with alignment that runs only when the

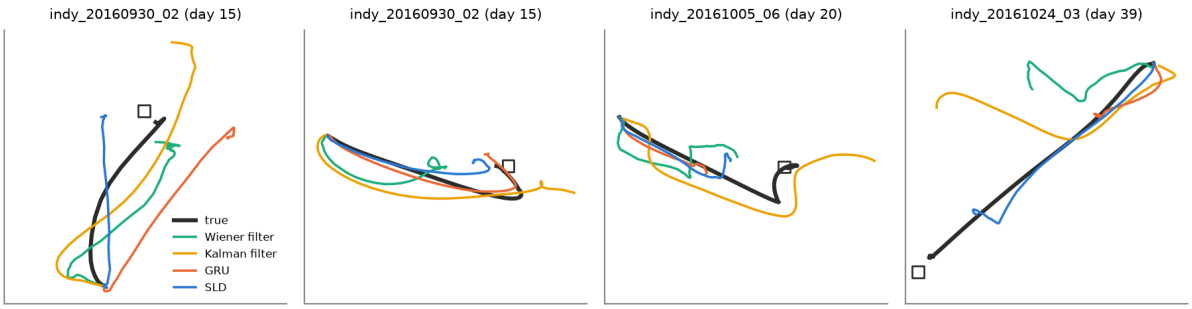


Figure 10: Held-out reaches: true cursor path (black) and the path obtained by integrating each decoder's velocity from the true start. Left two: last calibration day. Right two: later days, re-centred (Wiener, Kalman, GRU) or aligned (SLD), no labels.

detector fires, and with closed-loop checks before the adapted decoder is trusted. Alignment that runs unconditionally costs offline accuracy on ordinary days, and on one re-mapping day two of three seeds diverged. The closed-loop simulation adds a caveat in the other direction: on later days the aligned SLD adapter gave about three times the simulated bitrate of a fresh re-centred adapter (1.21 vs 0.41 bits/s) despite similar offline R^2, so the decision should be validated in closed loop, not offline.

Few labels, few parameters. When a few labelled reaches are available, the parameters to update matter more than the model class. Re-fitting everything (Wiener from scratch, full GRU fine-tuning) overfits one or five reaches; re-fitting only a day-specific input adapter, as POYO and NDT2 do with unit or context embeddings [20, 21], never hurts. This argues for keeping a small, explicitly day-specific component in deployed decoders even if it brings nothing when no labels are available.

Offline vs. closed loop. The simulator reproduces the textbook warning that offline accuracy and closed-loop performance diverge [4, 5]: a decoder with half the R^2 (Wiener) comes within reach of the best neural decoders after smoothing, ReFIT-style recalibration roughly doubles its bitrate, and the Kalman filter's offline weakness is not fixed by closed-loop relabelling. The simulator ranks decoders noisily (about ±0.2–0.4 bits/s with four 60 s sessions), so the closed-loop comparisons we trust are the large ones. A full-scale run (`configs/gpu.yaml`) uses ten sessions per condition.

The Kalman baseline. Our velocity Kalman filter is weak offline ($R^2 \approx 0.2$–0.3) because at 20 ms bins its single smoothing knob trades lag against noise; position-augmented states, larger bins or intention-trained (ReFIT) closed-loop fits are the standard fixes and would narrow the gap. We report it as tuned, not as engineered for this setting.

Distance to the product. The best simulated decoder reaches about a third of the perfect-decoder ceiling, and every decoder loses throughput on larger grids, so precision at rest, not speed, is the bottleneck for a 96-channel signal. Neuralink's implant records from 1,024 electrodes, and recent public reports describe pretraining on large neural datasets to shorten calibration; both are outside what can be tested on one public monkey dataset and a laptop CPU.

7 Limitations

- **Animal, arm, one array.** The spikes come from a monkey moving its arm, not a person with paralysis attempting movement, from a 96-electrode Utah array rather than Neuralink's 1,024-electrode threads. Real clinical nonstationarity (thread retraction) may be larger and different in kind.
- **Simulated closed loop.** The bitrates are produced by a model of the user and of the neural response to intention. Open-loop fidelity is checked (Fig. 6), but user learning, fatigue, attention and real click decoding are modelled, not measured. Absolute bitrates are not comparable to human Webgrid scores; rankings and trends are the output.
- **CPU scale.** Models are small (0.11 M parameters for the GRU, 0.17 M for SLD), trained for 1,500 steps (11–24 minutes each) with three seeds on a shared laptop CPU; few-shot curves use one seed on every fourth held-out day; ablations use one seed. Differences of a few hundredths of R^2 are within seed noise.
- **Single subject.** Cross-subject transfer and pretraining at scale [21, 22] are not tested.

8 Broader impact and safety

Cursor BCIs give people with paralysis back independent use of a computer; reliability across days is a safety and dignity issue as much as a performance one: a decoder that silently degrades produces unintended clicks.

Two practical lessons from this study apply to deployment. First, cheap label-free safeguards (per-channel re-centring) should be on by default and their effect monitored. Second, unsupervised adaptation can make things worse when its assumptions fail, so it needs a guard (here an identity penalty and a start from the re-centred decoder) and closed-loop monitoring. The simulator is a development tool for ranking decoders and catching closed-loop failures before user time is spent; it does not replace clinical evaluation. All data are public, de-identified animal recordings.

9 Reproducing at full scale

Everything here is produced by `scripts/run_cpu.sh` (CPU config `configs/cpu.yaml`). `scripts/run_gpu.sh` runs the same code with `configs/gpu.yaml` on a Linux CUDA machine (device auto-detected, mixed precision): 6,000 training steps with batch 128, five seeds, few-shot curves on every held-out day with five seeds, unsupervised alignment of every ablation on every day, and ten simulated WebGrid sessions per condition. Requirements are listed in `requirements-gpu.txt`; the script downloads the data from Zenodo. Results land in `results/`, and `python -m telepathy.report` regenerates the numbers and tables in this paper.

References

[1] Neuralink. Webgrid. https://neuralink.com/webgrid/. Accessed October 2026.

[2] Neuralink. PRIME study progress update. https://neuralink.com/blog/prime-study-progress-update/, 2024.

[3] Vikash Gilja, Paul Nuyujukian, Cindy A. Chestek, John P. Cunningham, Byron M. Yu, Joline M. Fan, Mark M. Churchland, Matthew T. Kaufman, Jonathan C. Kao, Stephen I. Ryu, and Krishna V. Shenoy. A high-performance neural prosthesis enabled by control algorithm design. *Nature Neuroscience*, 15 (12):1752–1757, 2012.

[4] John P. Cunningham, Paul Nuyujukian, Vikash Gilja, Cindy A. Chestek, Stephen I. Ryu, and Krishna V. Shenoy. A closed-loop human simulator for investigating the role of feedback control in brain-machine interfaces. *Journal of Neurophysiology*, 105(4):1932–1949, 2011.

[5] Francis R. Willett, Daniel R. Young, Brian A. Murphy, William D. Memberg, Christine H. Blabe, Chethan Pandarinath, Sergey D. Stavisky, Paymon Rezaii, Jad Saab, Benjamin L. Walter, Jennifer A. Sweet, Jonathan P. Miller, Jaimie M. Henderson, Krishna V. Shenoy, John D. Simeral, Beata Jarosiewicz, Leigh R. Hochberg, Robert F. Kirsch, and A. Bolu Ajiboye. Principled BCI decoder design and parameter selection using a feedback control model. *Scientific Reports*, 9:8881, 2019.

[6] Brianna M. Karpowicz, Joel Ye, Chaofei Fan, Pablo Tostado-Marcos, Fabio Rizzoglio, Clay Washington, Thiago Scodeler, Diogo de Lucena, Samuel R. Nason-Tomaszewski, Matthew J. Mender, Xuan Ma, Ezequiel Matias Arneodo, Leigh R. Hochberg, Cynthia A. Chestek, Jaimie M. Henderson, Timothy Q. Gentner, Vikash Gilja, Lee E. Miller, Adam G. Rouse, Robert A. Gaunt, Jennifer L. Collinger, and Chethan Pandarinath. Few-shot algorithms for consistent neural decoding (FALCON) benchmark. In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*, volume 37, 2024.

[7] Joseph E. O'Doherty, Mariana M. B. Cardoso, Joseph G. Makin, and Philip N. Sabes. Nonhuman primate reaching with multichannel sensorimotor cortex electrophysiology. Zenodo, 2017.

[8] David Sussillo, Sergey D. Stavisky, Jonathan C. Kao, Stephen I. Ryu, and Krishna V. Shenoy. Making brain-machine interfaces robust to future neural variability. *Nature Communications*, 7:13749, 2016.

[9] Joel Ye and Chethan Pandarinath. Representation learning for neural population activity with Neural Data Transformers. *Neurons, Behavior, Data analysis, and Theory*, 2021. arXiv:2108.01210.

[10] Brianna M. Karpowicz, Yahia H. Ali, Lahiru N. Wimalasena, Andrew R. Sedler, Mohammad Reza Keshtkaran, Kevin Bodkin, Xuan Ma, Daniel B. Rubin, Ziv M. Williams, Sydney S. Cash, Leigh R. Hochberg, Lee E. Miller, and Chethan Pandarinath. Stabilizing brain-computer interfaces through alignment of latent dynamics. *Nature Communications*, 16:4662, 2025. doi: 10.1038/s41467-025-59652-y. Preprint: bioRxiv 10.1101/2022.04.06.487388 (2022).

[11] Wei Wu, Yun Gao, Elie Bienenstock, John P. Donoghue, and Michael J. Black. Bayesian population decoding of motor cortical activity using a Kalman filter. *Neural Computation*, 18(1):80–118, 2006.

[12] Matthew S. Willsey, Samuel R. Nason-Tomaszewski, Scott R. Ensel, Hisham Temmar, Matthew J. Mender, Joseph T. Costello, Parag G. Patil, and Cynthia A. Chestek. Real-time brain-machine interface in non-human primates achieves high-velocity prosthetic finger movements using a shallow feedforward neural network decoder. *Nature Communications*, 13:6899, 2022.

[13] Paul Nuyujukian, Joline M. Fan, Jonathan C. Kao, Stephen I. Ryu, and Krishna V. Shenoy. A high-performance keyboard neural prosthesis enabled by task optimization. *IEEE Transactions on Biomedical Engineering*, 62(1):21–29, 2015.

[14] Chethan Pandarinath, Paul Nuyujukian, Christine H. Blabe, Brittany L. Sorice, Jad Saab, Francis R. Willett, Leigh R. Hochberg, Krishna V. Shenoy, and Jaimie M. Henderson. High performance communication by people with paralysis using an intracortical brain-computer interface. *eLife*, 6:e18554, 2017.

[15] Juan A. Gallego, Matthew G. Perich, Raeed H. Chowdhury, Sara A. Solla, and Lee E. Miller. Long-term stability of cortical population dynamics underlying consistent behavior. *Nature Neuroscience*, 23:260–270, 2020.

[16] Alan D. Degenhart, William E. Bishop, Emily R. Oby, Elizabeth C. Tyler-Kabara, Steven M. Chase, Aaron P. Batista, and Byron M. Yu. Stabilization of a brain-computer interface via the alignment of low-dimensional spaces of neural activity. *Nature Biomedical Engineering*, 4(7):672–685, 2020.

[17] Ali Farshchian, Juan A. Gallego, Joseph P. Cohen, Yoshua Bengio, Lee E. Miller, and Sara A. Solla. Adversarial domain adaptation for stable brain-machine interfaces. In *International Conference on Learning Representations (ICLR)*, 2019. arXiv:1810.00045.

[18] Chethan Pandarinath, Daniel J. O'Shea, Jasmine Collins, Rafal Jozefowicz, Sergey D. Stavisky, Jonathan C. Kao, Eric M. Trautmann, Matthew T. Kaufman, Stephen I. Ryu, Leigh R. Hochberg, Jaimie M. Henderson, Krishna V. Shenoy, L. F. Abbott, and David Sussillo. Inferring single-trial neural population dynamics using sequential auto-encoders. *Nature Methods*, 15 (10):805–815, 2018.

[19] Beata Jarosiewicz, Anish A. Sarma, Daniel Bacher, Nicolas Y. Masse, John D. Simeral, Brittany Sorice, Erin M. Oakley, Christine Blabe, Chethan Pandarinath, Vikash Gilja, Sydney S. Cash, Emad N. Eskandar, Gerhard Friehs, Jaimie M. Henderson, Krishna V. Shenoy, John P. Donoghue, and Leigh R. Hochberg. Virtual typing by people with tetraplegia using a self-calibrating intracortical brain-computer interface. *Science Translational Medicine*, 7(313):313ra179, 2015.

[20] Joel Ye, Jennifer L. Collinger, Leila Wehbe, and Robert Gaunt. Neural data transformer 2: Multi-context pretraining for neural spiking activity. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36, 2023.

[21] Mehdi Azabou, Vinam Arora, Venkataramana Ganesh, Ximeng Mao, Santosh Nachimuthu, Michael J. Mendelson, Blake Richards, Matthew G. Perich, Guillaume Lajoie, and Eva L. Dyer. A unified, scalable framework for neural population decoding. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36, 2023. arXiv:2310.16046.

[22] Joel Ye et al. A generalist intracortical motor decoder. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 38, 2025. bioRxiv 10.1101/2025.02.02.634313.

[23] Chaofei Fan, Nick Hahn, Foram Kamdar, Donald Avansino, Guy H. Wilson, Leigh Hochberg, Krishna V. Shenoy, Jaimie M. Henderson, and Francis R. Willett. Plug-and-play stability for intracortical brain-computer interfaces: A one-year demonstration of seamless brain-to-text communication. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36, 2023. arXiv:2311.03611.

[24] Joseph G. Makin, Joseph E. O'Doherty, Mariana M. B. Cardoso, and Philip N. Sabes. Superior arm-movement decoding from cortex with a new, unsupervised-learning algorithm. *Journal of Neural Engineering*, 15(2):026010, 2018.

[25] Felix Pei, Joel Ye, David Zoltowski, Anqi Wu, Raeed H. Chowdhury, Hansem Sohn, Joseph E. O'Doherty, Krishna V. Shenoy, Matthew T. Kaufman, Mark Churchland, Mehrdad Jazayeri, Lee E. Miller, Jonathan Pillow, Il Memming Park, Eva L. Dyer, and Chethan Pandarinath. Neural latents benchmark '21: Evaluating latent variable models of neural population activity. In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*, 2021. arXiv:2109.04463.